EX No: 3

Classification with Decision Trees

Code:

```
# Step 1: Import Libraries
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, accuracy_score
import matplotlib.pyplot as plt
import seaborn as sns


# Step 2: Load Dataset
iris = load_iris()
X = iris.data
y = iris.target


# Step 3: Split the dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)


# Step 4: Train the Decision Tree Classifier
dt_model = DecisionTreeClassifier(criterion='gini', random_state=0)
dt_model.fit(X_train, y_train)

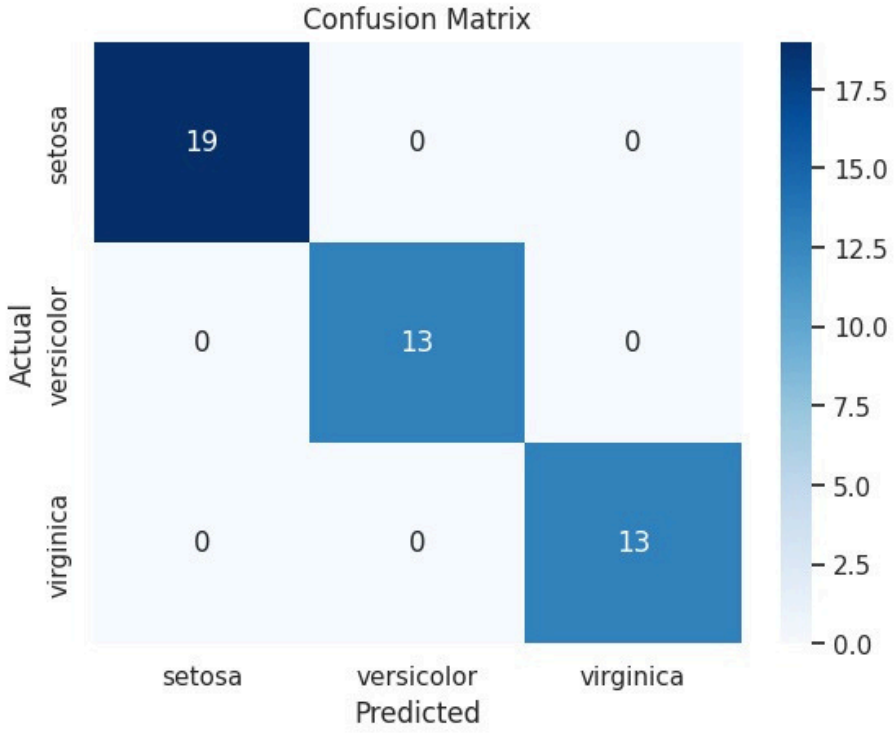
# Step 5: Predict
y_pred = dt_model.predict(X_test)
```

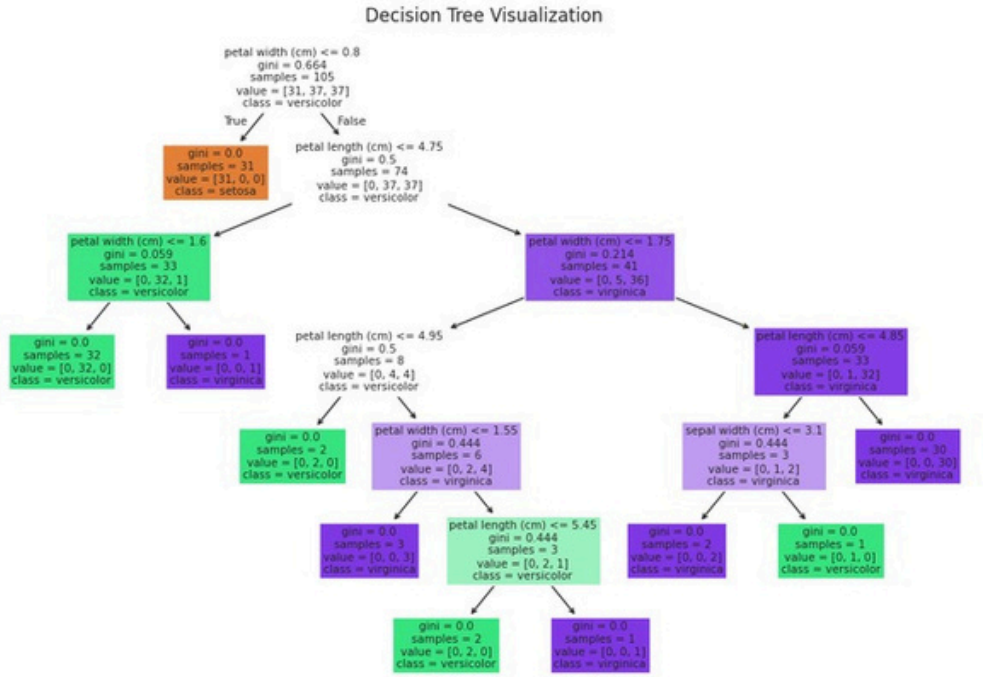
```
# Step 6: Evaluate the Model
cm = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)
print("Confusion Matrix:\n", cm)
print("Accuracy Score:", acc)


# Step 7: Visualize Confusion Matrix
sns.heatmap(cm, annot=True, cmap="Blues",
xticklabels=iris.target_names,
yticklabels=iris.target_names)
plt .xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt .show()
```

Output:

```
Confusion Matrix:
 [[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
Accuracy Score: 1.0
```


Confusion Matrix
setosa
versicolor
virginica
Actual
19
0
0
0
13
0
0
0
13
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
setosa
versicolor
virginica
Predicted


Decision Tree Visualization
petal width (cm) <= 0.8
gini = 0.664
samples = 105
value = [31, 37, 37]
class = versicolor
True
False
gini = 0.0
samples = 31
value = [31, 0, 0]
class = setosa
petal length (cm) <= 4.75
gini = 0.5
samples = 74
value = [0, 37, 37]
class = versicolor
petal width (cm) <= 1.6
gini = 0.059
samples = 33
value = [0, 32, 1]
class = versicolor
petal width (cm) <= 1.75
gini = 0.214
samples = 41
value = [0, 5, 36]
class = virginica
gini = 0.0
samples = 32
value = [0, 32, 0]
class = versicolor
gini = 0.0
samples = 1
value = [0, 0, 1]
class = virginica
petal length (cm) <= 4.95
gini = 0.5
samples = 8
value = [0, 4, 4]
class = versicolor
petal length (cm) <= 4.85
gini = 0.059
samples = 33
value = [0, 1, 32]
class = virginica
gini = 0.0
samples = 2
value = [0, 2, 0]
class = versicolor
petal width (cm) <= 1.55
gini = 0.444
samples = 6
value = [0, 2, 4]
class = virginica
sepal width (cm) <= 3.1
gini = 0.444
samples = 3
value = [0, 1, 2]
class = virginica
gini = 0.0
samples = 30
value = [0, 0, 30]
class = virginica
gini = 0.0
samples = 3
value = [0, 0, 3]
class = virginica
petal length (cm) <= 5.45
gini = 0.444
samples = 3
value = [0, 2, 1]
class = versicolor
gini = 0.0
samples = 2
value = [0, 0, 2]
class = virginica
gini = 0.0
samples = 1
value = [0, 1, 0]
class = versicolor
gini = 0.0
samples = 2
value = [0, 2, 0]
class = versicolor
gini = 0.0
samples = 1
value = [0, 0, 1]
class = virginica